

		answer
5(a)(i)	Work function energy is the minimum amount of energy needed to liberate an electron from the metal surface.	[1] for correct answer
5(a)(ii)	The photons each of same magnitude of energy ($E=hf$) are received by the electrons in the metal surface. The electrons nearest to the metal surface require the least amount of energy to be emitted as photoelectrons. Electrons deeper under the surface of the metal require more energy to be emitted. Hence, the emitted photoelectrons have a range of kinetic energies up to a maximum. <i>Comments:</i> - many omitted the explanation for maximum kinetic energy of the emitted photoelectrons. - there is the need to make reference to the fact that each photon of the monochromatic radiation carries the same, discrete amount of energy.	[1] for correct answer [1] for correct answer
5(a)(iii)	Using Einstein's photoelectric equation, $\frac{hc}{\lambda} = \Phi + KE_{\max}$,	[1] for correct expression and numerical substitution [1] for correct answer

No.	Solution	Remarks
	$\lambda = \frac{hc}{\Phi + KE_{\max}}$ $= \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{(4.07 \times 1.60 \times 10^{-19}) + 3.65 \times 10^{-20}}$ $= 289.22 \times 10^{-9} \text{ m}$ $\approx 2.89 \times 10^{-7} \text{ m}$	
5(b)(i)	Time taken $= \frac{4.07 \times 1.60 \times 10^{-19}}{3.0 \times 10^{-22}}$ $= 2170.7$ $\approx 2170 \text{ s}$	[1] for correct answer
5(b)(ii)	Upon incident on the metal surface, the photons undergo 1 to 1 interaction with the electrons. The energy is transferred instantaneously to the electron which accounts for the instantaneous emission of electrons as photoelectrons without time lag. Hence, this interaction shows that electromagnetic radiation travels as photons which are discrete packets of energy. <i>Comments:</i> - many could not explain the experimental evidence for the particulate nature of electromagnetic radiation. - the concept of 1 photon giving all its discrete amount of energy to only 1 electron, in one-to-one interaction, was not well understood.	[1] for correct answer [1] for correct answer
6(a)	Isotopes are nuclides with the same number of protons but	[1] for correct